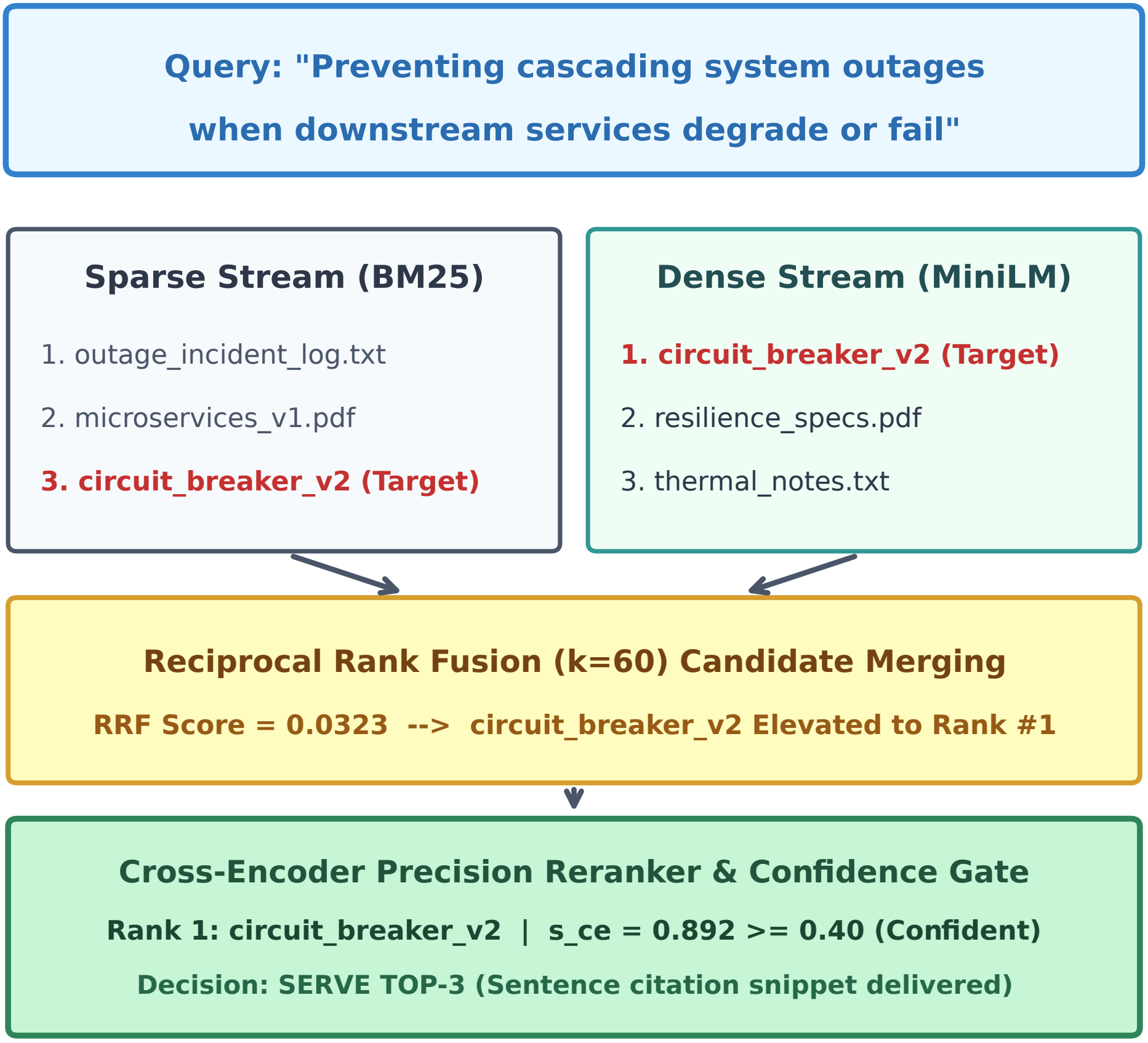
Query: "Preventing cascading system outages when downstream services degrade or fail"
Sparse Stream (BM25)
1. outage_incident_log.txt
2. microservices_v1.pdf
3. circuit_breaker_v2 (Target)
Dense Stream (MiniLM)
1. circuit_breaker_v2 (Target)
2. resilience_specs.pdf
3. thermal_notes.txt
Reciprocal Rank Fusion (k=60) Candidate Merging
RRF Score = 0.0323 --> circuit_breaker_v2 Elevated to Rank #1
Cross-Encoder Precision Reranker & Confidence Gate
Rank 1: circuit_breaker_v2 | s_ce = 0.892 >= 0.40 (Confident)
Decision: SERVE TOP-3 (Sentence citation snippet delivered)